

Opinion Network Formation: Extended Technical Report

DPCN Assignment 1 (Deep-Dive Companion to the 8-Page Submission)

Team 34

Team Name

Team 34

GitHub link to your code

<https://github.com/vishakkashyapk30/dpcn-a1-opinion-network-team34>

Dataset Documentation

Our class survey asks 60 opinion statements, each answered on a five-point Likert scale: Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree. The 60 statements are grouped into four themes, 15 items each: **Technology (T)**, **Education (E)**, **Ethics/Society (S)**, and **Environment (V)**. The raw file has 96 respondents.

We mapped the five labels onto a symmetric number line, so a computer can compare answers numerically:

Label	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Code	−1.0	−0.5	0.0	+0.5	+1.0

Blank cells and the literal text “No Comments” were coded as missing, not as Neutral. Coding a skipped question as Neutral would invent an opinion the respondent never gave, and it would make two people who both skipped the same questions look artificially similar later on. We kept only respondents who answered at least 50% of the 60 items: **87 of 96 kept, 9 dropped** (ids 30, 39, 44, 60, 68, 73, 77, 78, 87). Missing answers are also not spread evenly across themes: Technology has the fewest missing cells and Environment the most, $T < E < S < V$ in order. We return to what that pattern might mean in the Results and Discussion section.

The whole pipeline treats this table as a **bipartite** structure: respondents on one side, statements on the other, with a link wherever someone answered a statement. Squashing away the statements and keeping only respondents gives an **agent network** (who thinks like whom). Squashing away the respondents and keeping only statements gives an **attitude network** (which beliefs travel together). Both are projections of the same underlying survey, following MacCarron, Maher and Quayle (2020) and Dinkelberg et al. (2021), with applied framing from Maher, MacCarron and Quayle (2020).

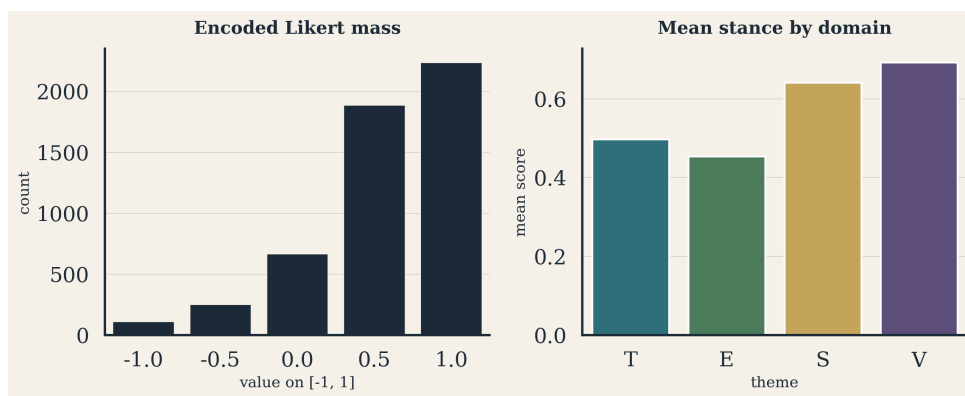


Figure 1: Encoded response mass and mean stance by theme. Left: how many answers fell at each of the five values on the $[-1, 1]$ scale. Right: the average score per theme (0 = perfectly neutral).

The left panel shows a clear Agree skew: few respondents pick Strongly Disagree, progressively more pick Disagree, Neutral, Agree, with Agree and Strongly Agree the two largest buckets. The right panel shows every theme's mean sits above zero, with Environment and Ethics/Society highest and Education lowest.

Pipeline Followed

This section explains, step by step and in plain language, every algorithm used to turn the 87x60 opinion table into the two networks and the analysis on top of them.

Step 1: Measuring how similar two respondents are

For any two respondents i and j , we only compare them on items both answered: call that shared set F_{ij} , with $n_{ij} = |F_{ij}|$ items in it. For every shared item f , their scores q_{if} and q_{jf} differ by $|q_{if} - q_{jf}|$, between 0 (identical answer) and 2 (one Strongly Agree, the other Strongly Disagree). We sum that disagreement across every shared item and subtract it from the number of shared items:

$$S_{ij} = n_{ij} - \sum_{f \in F_{ij}} |q_{if} - q_{jf}|$$

Worked example: suppose two people both answered the same 4 items, with scores (1.0, 0.5, -0.5, 1.0) for person A and (1.0, 0.0, -1.0, 0.5) for person B. The differences are 0, 0.5, 0.5, 0.5, summing to 1.5, so $S_{ij} = 4 - 1.5 = 2.5$. Perfect agreement on every item would give $S_{ij} = n_{ij} = 4$; exact opposite answers on everything would give $S_{ij} = -4$. So higher S_{ij} means stronger agreement, and the score is automatically scaled by how many items the pair actually shares.

We only trust S_{ij} when $n_{ij} \geq 20$, to avoid a similarity score driven by a handful of shared items. In this dataset that floor never actually binds: every one of the $\binom{87}{2} = 3,741$ pairs shares at least 41 of the 60 items, well above 20, because the incompleteness filter had already removed the very sparse respondents. Across all pairs, S_{ij} ranges from -3 to 52 (mean 33.7, median 34.5).

Step 2: Choosing an agreement threshold

S_{ij} alone gives a complete network: every pair has some score, so drawing an edge for every pair produces one dense blob with no visible structure. We only keep edges where $S_{ij} \geq \theta$ for a cutoff θ . Too high a threshold and only near-perfect agreement survives, shattering the graph into tiny disconnected pieces. Too low a threshold and almost every pair survives, making any detected community meaningless.

Following Dinkelberg et al. (2021), we scan θ downward from the highest observed similarity and stop where the network's **giant component** (the single largest group of respondents who can all reach each other through some chain of surviving edges) covers about 80% of everyone.

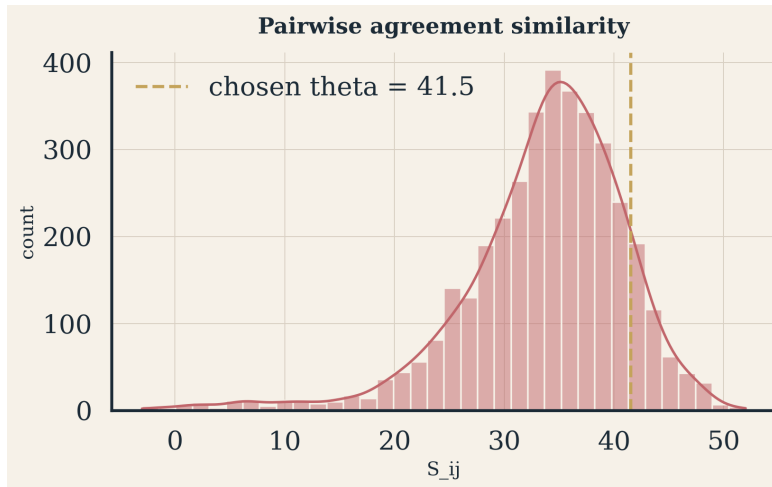


Figure 2: Distribution of pairwise agreement scores S_{ij} across all 3,741 respondent pairs. The dashed line marks the threshold we chose.

The histogram is roughly bell-shaped, centred a little above 30. The chosen threshold, $\theta = 41.5$, sits in the right-hand tail, above the 75th percentile of all pairwise scores (38.5). Only the top quarter or so of agreement pairs survive as edges.

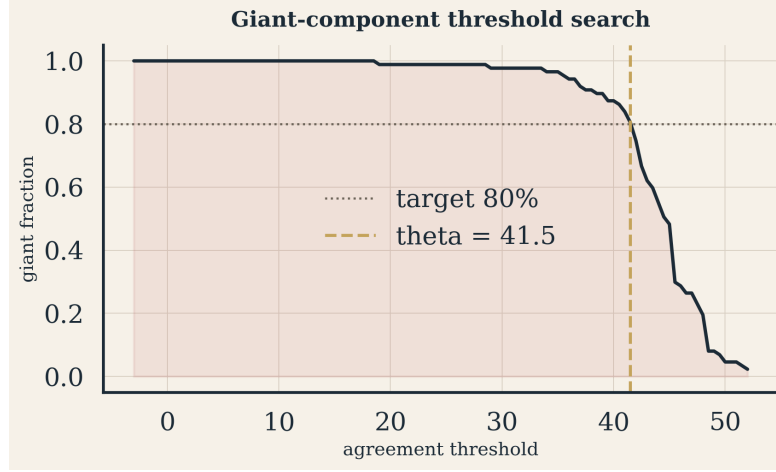


Figure 3: Giant-component fraction as the threshold is lowered from strict to lenient. The horizontal dashed line is the 80% target; the vertical dashed line is the chosen θ .

As θ decreases, more edges are added and the giant component grows in an S-shaped curve: near zero when only extreme agreement counts, climbing steeply through a middle range, then flattening near 1.0. We stop the first time this curve crosses 80%, at $\theta = 41.5$, a single pre-committed stopping rule rather than a threshold picked to produce a nice-looking picture.

Step 3: Building the agent (respondent) network

At $\theta = 41.5$:

Property	Value	Plain-English meaning
Nodes	87	every kept respondent, including disconnected ones
Edges	456	pairs whose agreement cleared the bar
Density	0.122	12.2% of all possible respondent pairs are connected
Average degree	10.48	a typical respondent shares a strong-agreement tie with about 10 to 11 others
Average clustering	0.447	see below
Giant component	70 / 87 nodes (80.5%)	matches our 80% target by construction
Avg. shortest path (giant only)	2.27	see below

Average clustering asks: if A agrees strongly with both B and C, do B and C also agree strongly with each other? A value of 0.447 is fairly high (a random graph with the same density would sit near 0.12), meaning agreement here forms cohesive little neighbourhoods rather than spreading out evenly. Average shortest path length asks: inside the giant component, how many agreement-edges do you need to hop across to connect any two people, on average? At 2.27, most well-connected respondents are only one or two hops from each other, even though the graph overall is fairly sparse.

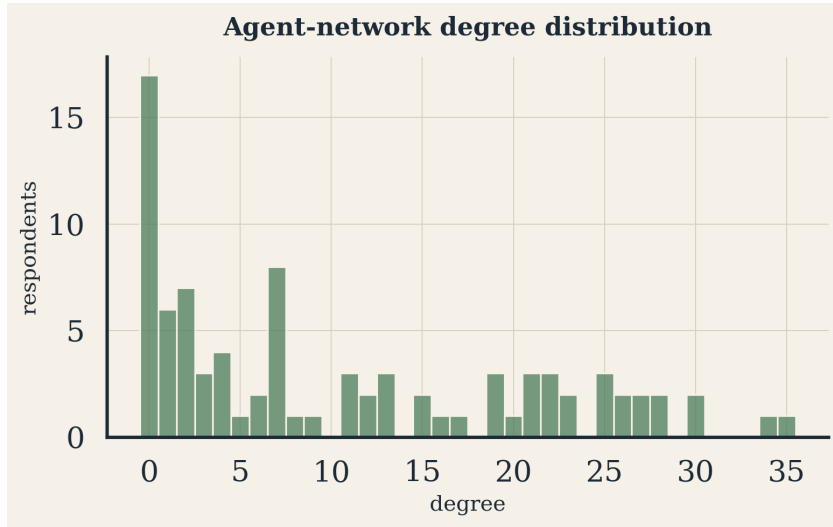


Figure 4: Agent-network degree distribution: how many respondents have each degree value.

There is a tall spike of 17 respondents with degree exactly zero: people who cleared the completeness filter but whose answer pattern did not agree strongly enough with anyone else to earn an edge at this threshold. The rest of the distribution spreads broadly from about 1 up to 35, with no small set of super-connected hubs.

Step 4: Building the attitude (statement) network

For two statements a and b , define a co-endorsement weight

$$W_{ab} = \sum_i q_{ia} q_{ib}$$

summed over every respondent who answered both. The product of two signed scores is positive exactly when both point the same way (both agreed, or both disagreed) and grows with how strongly both were held. Summing over 87 respondents gives, in total, how much the whole class endorses both statements together. We keep only edges above the 70th percentile of positive W_{ab} values ($W \geq 40.75$):

Property	Value
Nodes	60 (every statement)
Edges	498
Density	0.281
Average degree	16.6
Average clustering	0.606
Giant component	44 / 60 nodes (73.3%)
Avg. shortest path (giant only)	1.48

The attitude network is denser and more tightly clustered than the agent network (0.606 vs 0.447 clustering), meaning statements that travel together do so very consistently.

Step 5: Finding groups with two different algorithms

Louvain (bottom-up). Start with every node in its own community. Try moving each node into a neighbouring community and keep the move if it increases **modularity**, a score that asks whether edges inside proposed communities are denser than random chance would predict given each node's degree. Repeat until no move helps, then collapse each community into a single super-node and repeat one level up. This is a greedy method built up from individual nodes, not a top-down split.

Running Louvain (weighted by S_{ij} , fixed seed) gives modularity $Q = 0.275$: real but moderate structure, not sharply separated communities. There are 20 “communities” in total, but only three have more than one member: sizes 37, 29, and 4. The other 17 are singletons, and they are exactly the same 17 respondents with zero degree above. Louvain cannot merge a node with no edges into anyone’s group, so it correctly leaves each one alone, a useful cross-check between two independent calculations.

Girvan-Newman (top-down). Start with the whole giant component as one group. Compute **edge betweenness** for every edge (how many shortest paths between all other pairs pass through it) and delete the single edge with the highest betweenness, since it is the most likely bridge holding two regions together. Recompute and repeat until the graph splits. We stop at the first meaningful split.

Applied to the 70-node giant component, the first split is 64 / 4 / 2, heavily lopsided, unlike Louvain’s near-even 37/29 split of the whole graph. If the class held two roughly equal, opposed camps, Girvan-Newman would find a far more balanced cut. Instead it finds one dominant backbone with a couple of small splinter groups attached by thin bridges. Together with the moderate Louvain modularity, both algorithms point the same way: one broad shared core with graded internal variation, not a class split into two hostile camps.

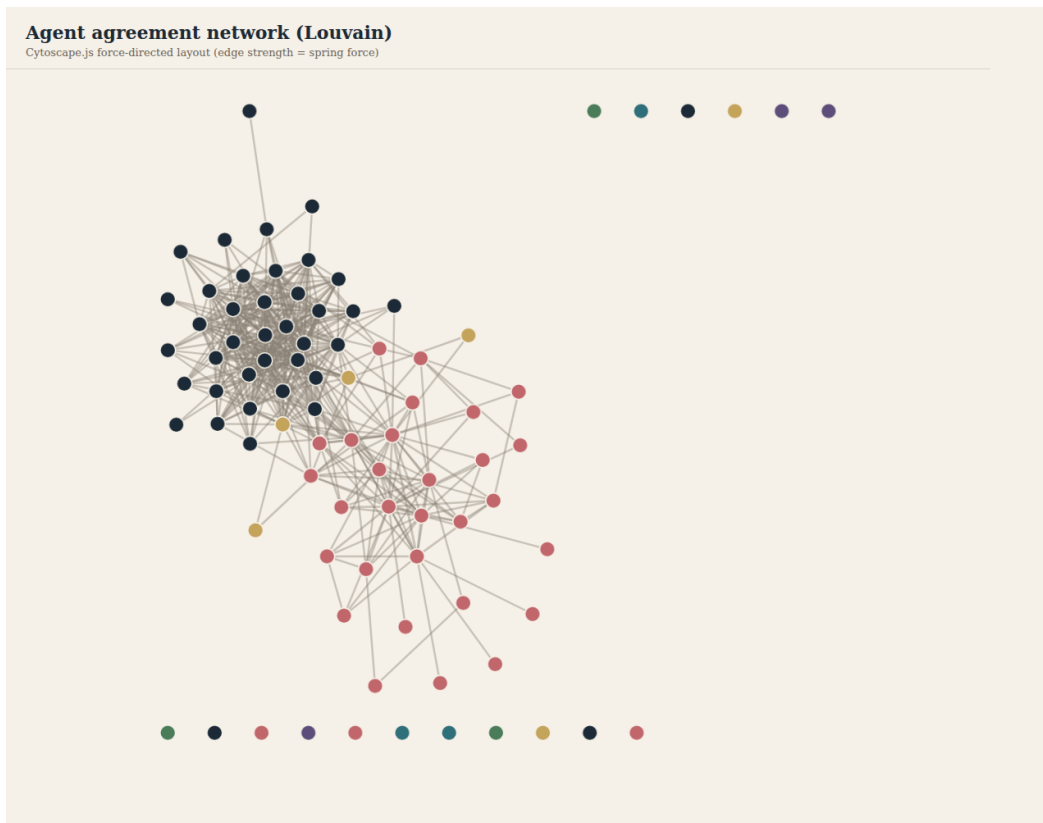


Figure 5: Cytoscape view of the agent network, nodes coloured by Louvain community. Interactive version: [open in browser](#).

One dense cluster on the left (size 37) sits directly attached to and interleaved with a second, looser cluster (size 29), with real edges connecting individual members across the two, not two floating islands. The size-4 community sits right at the seam. Disconnected singleton respondents are scattered at the very edges of the frame.

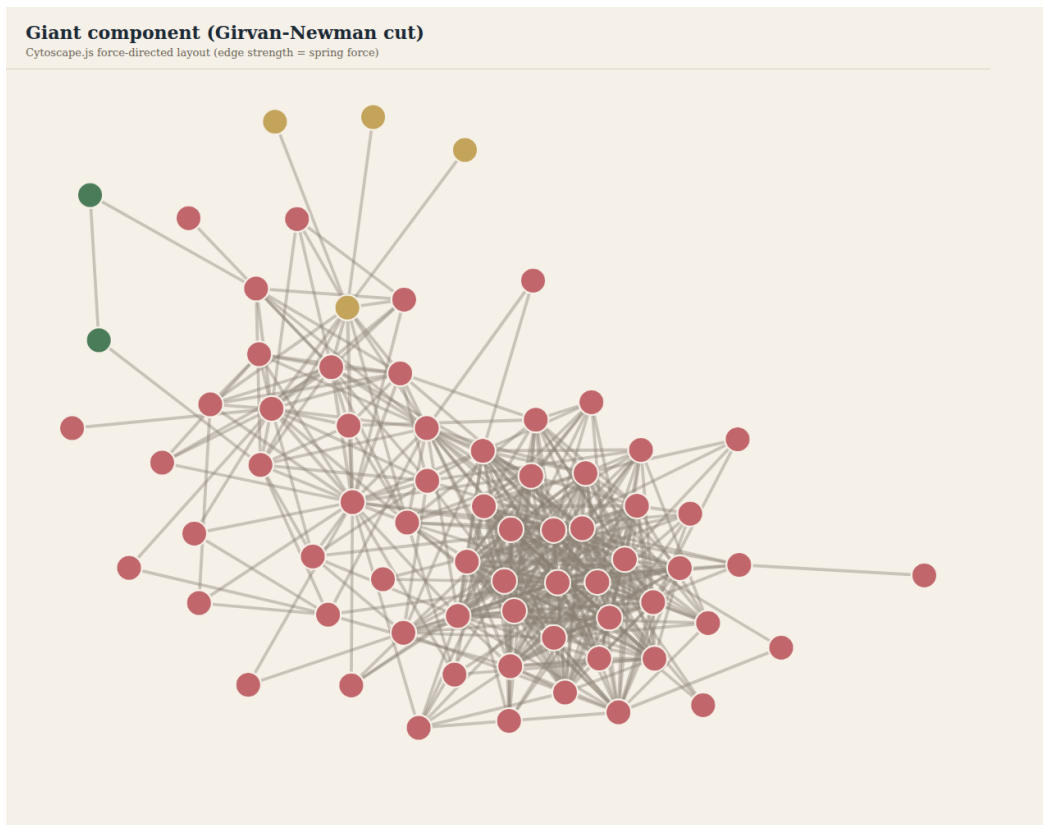


Figure 6: Giant component only, coloured by the Girvan-Newman split. Interactive version: [open in browser](#).

The overwhelming majority of nodes sit inside one dense mass. A very small number of nodes are tethered to that mass by only one or two long, thin edges, the small splinter groups Girvan-Newman peeled off first, because those edges had the highest betweenness. This is a backbone-and-fringe shape, not a symmetric two-sided divide.

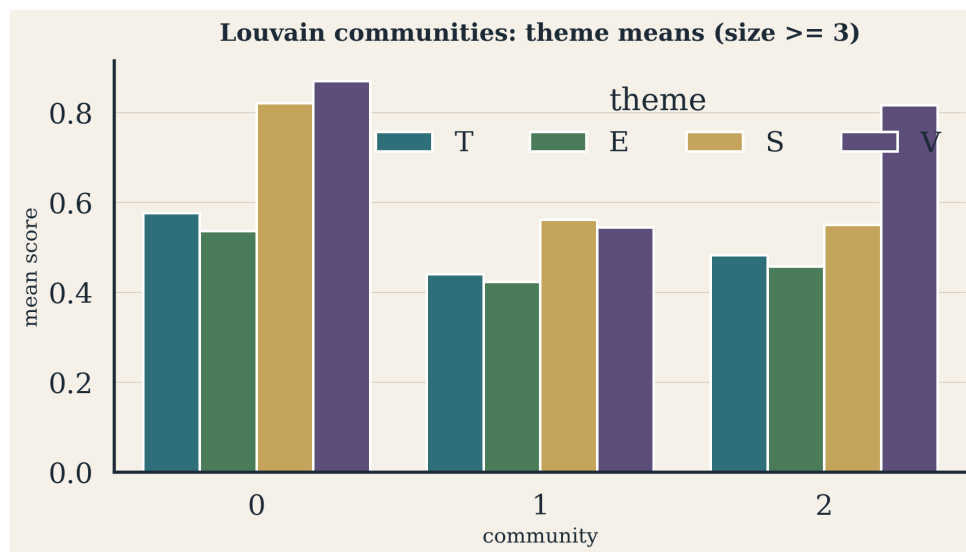


Figure 7: Louvain communities of size ≥ 3 : mean score on each of the four themes.

Both large communities are positive on every theme, so this is not an “agree vs disagree” split. Community 0 (size 37) sits noticeably higher than Community 1 (size 29) on every theme, especially Ethics/Society and Environment.

Education is the lowest bar in every community, a shared soft spot rather than a dividing line. The two big communities differ mainly in how strongly and how consistently they agree, not in what they agree about.

Step 6: Who is central, and by which definition?

A network has more than one sensible notion of importance. We compute three, on two different networks.

Degree centrality (agent network), who is a typical opinion-neighbour and who is an outlier. The 10 highest-degree respondents (ids 114, 67, 104, 55, 49, 110, 83, 61, 94, 105, degrees from 35 down to 26) overlap strongly with a large fraction of the class. The 17 zero-degree respondents are not all extreme in their own opinions: their individual mean scores range from -0.11 up to $+0.68$, median around $+0.41$, similar to the class median. Being disconnected at this threshold reflects an unusual pattern across items more than an unusually strong or weak overall opinion.

Eigenvector centrality (attitude network), which statements sit at the well-connected heart of the belief structure. A node's score is high if it connects to other highly-connected nodes, computed by repeatedly updating each node's score as the average of its neighbours' scores until it stops changing (power iteration). Top items: E15 (continuous learning throughout one's career, 0.230), V13 (companies accountable for environmental impact, 0.225), V09 (design products for reuse and recycling, 0.219), E11 (update curricula to match emerging tech, 0.219), S05 (individuals should control their personal data, 0.219).

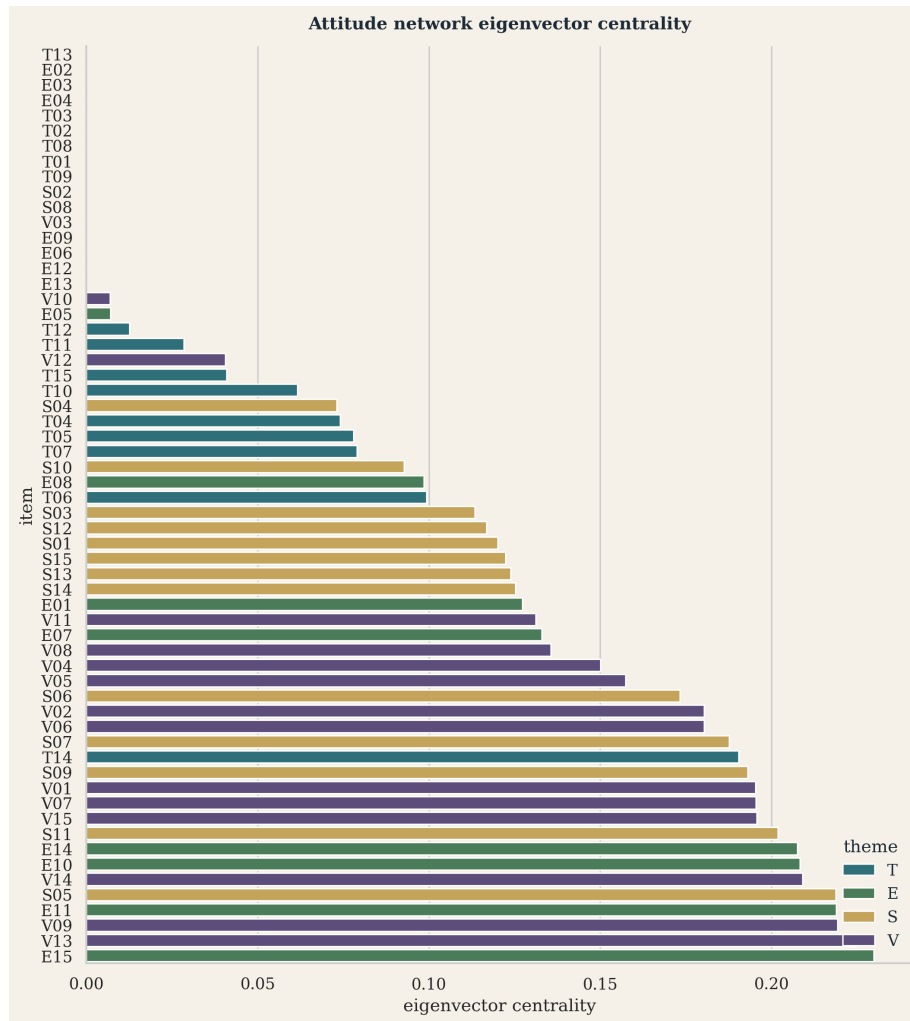


Figure 8: Eigenvector centrality of every attitude-network item, bars coloured by theme, sorted from least to most central.

The bottom of this chart, near-zero centrality, is almost entirely Technology and Education items (T13, E02, E03,

E04, T03, T02, T08, T01). The top is dominated by Environment and Ethics/Society items, with a handful of Education items (E15, E11, E14, E10) earning their way in through a “modernise the university” framing rather than a “how should a classroom be run” framing.

Betweenness centrality (attitude network), which statements bridge otherwise separate parts of the belief structure. A high score means removing that node would make many other pairs harder to connect through the network. Top items: T14 (social media algorithms cause polarization, betweenness 0.0452), V13 (0.0301), E15 (0.0245), V07 (protecting biodiversity, 0.0245), S11 (free expression short of harm or discrimination, 0.0212).

V13, E15, V07 and S11 score high on both betweenness and eigenvector centrality: deeply embedded hubs and bridges at once. T14 is different: its betweenness is nearly 50% higher than the runner-up, the single most important bridge in the network, yet its eigenvector score (0.190) is noticeably lower than the other top-betweenness items. T14, a Technology statement about social media and polarization, is not itself part of the densely-interconnected core, but it is the item that most connects the more fragmented Technology side of the survey to that core.

Step 7: Which statements organise the class into groups?

Two different questions are easy to conflate: which statements have the most disagreement (variance), versus which statements, if removed, would most change who ends up grouped with whom (structural importance).

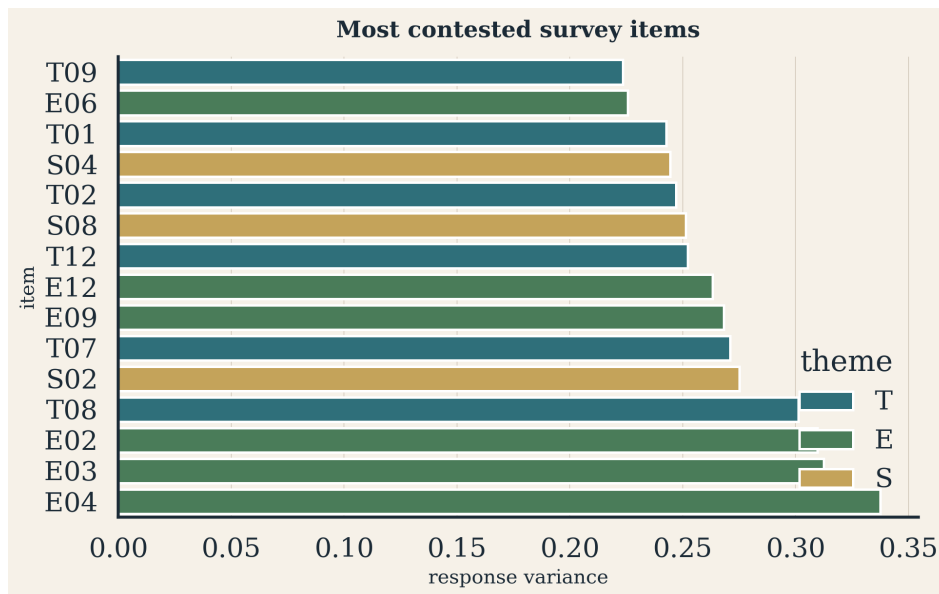


Figure 9: Top 15 items ranked by response variance.

Education items dominate the top: E04 (online learning can complement classroom teaching), E03 (attendance should be compulsory), E02 (exams accurately measure knowledge), alongside Technology items like T08 (AI-assisted diagnosis) and T07 (robots replacing humans in hazardous jobs). These are statements where the class is genuinely divided.

For structural importance, we shuffle one item’s answers randomly across respondents, breaking any real relationship between that item and everything else while keeping its own overall distribution unchanged. We rebuild the similarity network and Louvain communities on the shuffled data and compare the new labels to the original using the **Adjusted Rand Index (ARI)**, a score from about -1 to 1 where 1 means identical groupings and 0 means no better than random labelling. Importance is $1 - \overline{\text{ARI}}$ over three shuffle repeats: if shuffling an item barely moves the grouping, importance is low; if it causes the grouping to fall apart, importance is high.

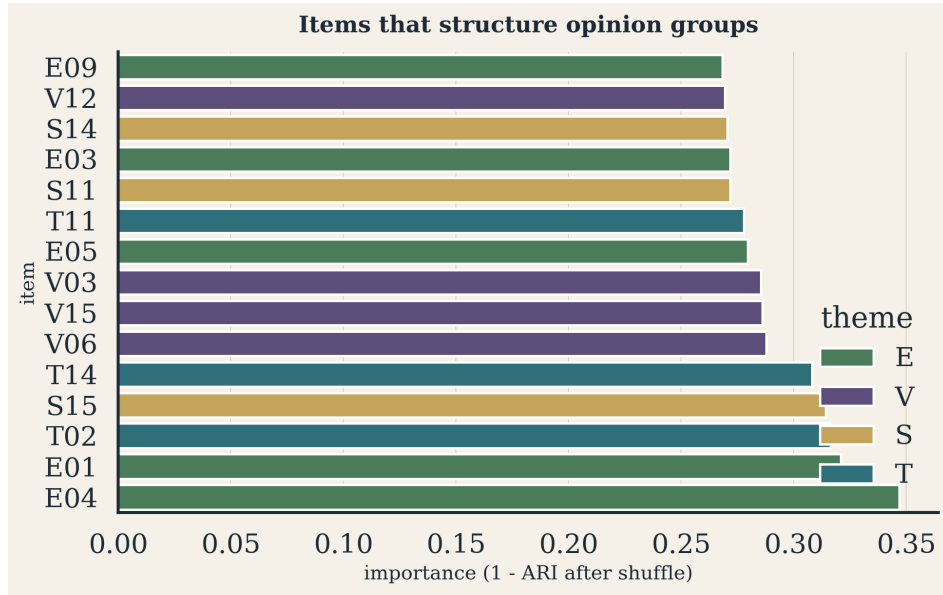


Figure 10: Top 15 items by structural importance ($1 - \overline{\text{ARI}}$ after shuffling).

E04 and E01 top the list, followed by T02 (generative AI as a learning aid), S15 (public trust in emerging technology), and T14, the same item that was the top betweenness bridge above, now confirmed independently as one of the items that most shapes who gets grouped with whom. Several Environment items (V06, V15, V03) also appear despite Environment's high mean scores: even where most people lean the same direction, small consistent differences in how strongly they lean can still separate the two large communities. E04, E03, T08, S02, E09 and E12 appear on both the variance and the importance lists, doubly interesting items that are both contested and structurally load-bearing.

Step 8: What ideas travel together

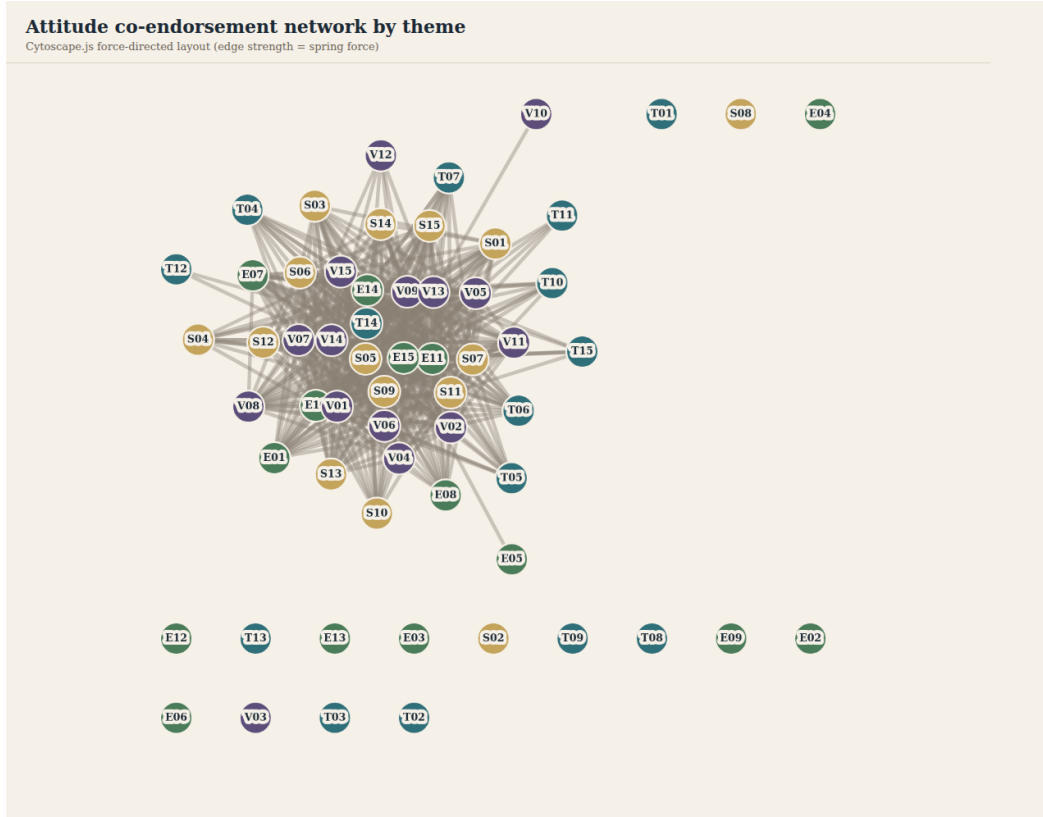


Figure 11: Attitude co-endorsement network, nodes coloured by theme. Interactive version: [open in browser](#).

Environment (purple) and Ethics/Society (gold) nodes sit thick and richly interconnected in the main mass. Education (green) contributes some core members too, mostly the “innovation and lifelong learning” items (E10, E11, E14, E15), but also has members sitting more loosely on the fringe. Technology (teal) is most often pushed to the edges or cut off from the core entirely.

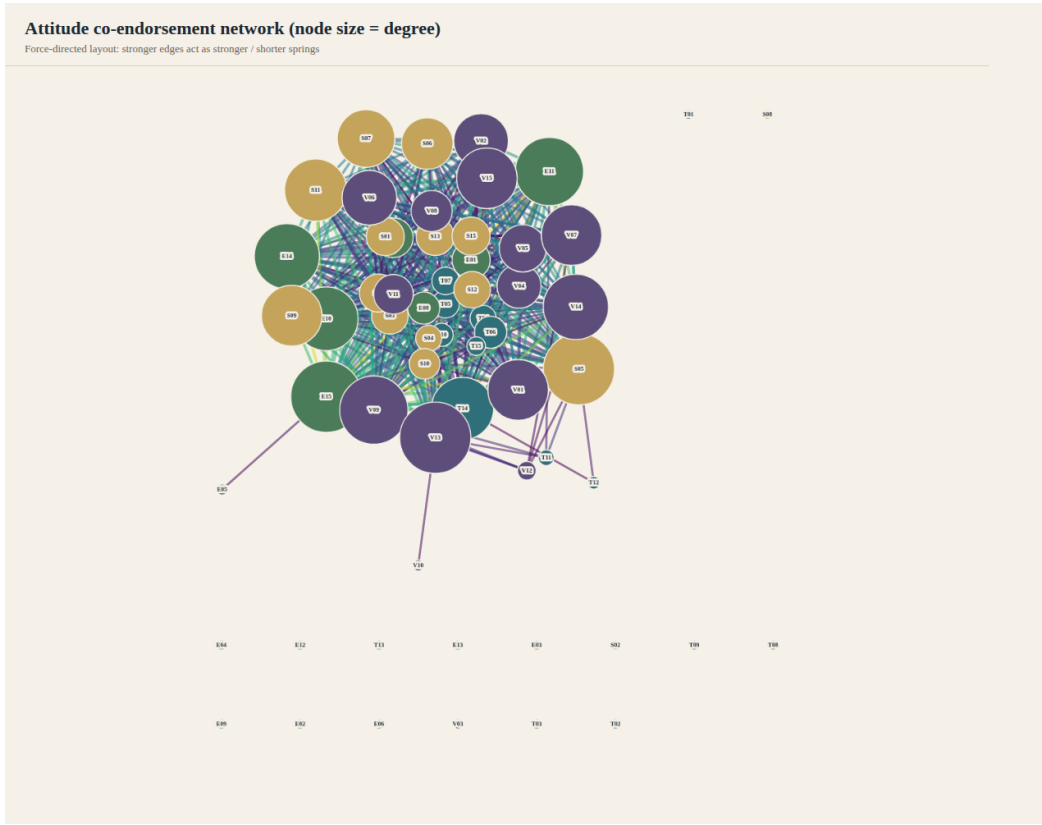


Figure 12: The same attitude network with node size scaled by degree, force-directed layout. Interactive version: [open in browser](#).

This view makes the core and periphery impossible to miss: a large, tightly-packed ball of big nodes in the centre-left, mostly gold (Ethics/Society) and purple (Environment) with a few large green (Education) nodes woven in, surrounded by many small, completely isolated single dots (T01, S08, E04, E12, T13, E13, S02, T09, T08, E09, E02, E06, V03, T03, T02). Every one of those isolated dots is a Technology or Education item. This is the clearest picture in the report of a “belief packaging” pattern: sustainability and social-responsibility statements bind tightly into one endorsed package, while several technology and education statements stand apart from that package and from each other.

The twenty most tightly bound pairs of statements

The co-endorsement weight W_{ab} used to build the attitude network gives us a ranked list of every pair of statements, so we can also ask a very direct question: out of all $\binom{60}{2} = 1,770$ possible pairs of statements, which specific 20 pairs does the class endorse together most strongly?

Table 4: E11 update curricula for emerging technology. E15 continuous learning throughout one’s career. E14 university-industry collaboration. E10 prioritize innovation over rote learning. V13 companies accountable for environmental impact. V09 design products for reuse and recycling. V14 international cooperation on environmental challenges. V15 protect natural resources for future generations. S05 individuals should control their personal data. S11 free expression short of harm or discrimination.

Rank	Pair	Weight	Rank	Pair	Weight
1	E11 x E15	63.25	11	E11 x S05	59.50

Rank	Pair	Weight	Rank	Pair	Weight
2	E15 x V13	62.00	12	E11 x V09	59.50
3	E15 x S05	61.75	13	V13 x V14	59.00
4	E14 x E15	61.50	14	E11 x V13	59.00
5	V09 x V13	61.25	15	S05 x V09	58.50
6	E15 x V09	61.00	16	E10 x V13	58.25
7	S05 x V13	60.50	17	V09 x V14	58.00
8	E10 x E15	60.50	18	E11 x S11	57.75
9	E15 x S11	60.00	19	E15 x V15	57.50
10	E15 x V14	59.50	20	E10 x V09	57.25

Three patterns in this table are worth pulling out on their own.

E15 is a mega-hub, not just a top item. It appears in 9 of these 20 pairs, more than any other statement, and it pairs with items from every one of the other three themes (E10/E11/E14 in Education, V09/V13/V14/V15 in Environment, S05/S11 in Ethics/Society). This is the same item that topped the eigenvector centrality ranking earlier, and this table shows exactly why: “continuous learning and skill development are essential throughout one’s career” is worded broadly and inoffensively enough that whoever strongly agrees with it also tends to strongly agree with almost everything else the class broadly endorses. It functions less like one specific belief and more like a proxy for general agreeableness, worth keeping in mind whenever E15’s high centrality is quoted elsewhere in this report.

Not one Technology item appears anywhere in the top 20. Out of 1,770 possible pairs, the 20 strongest are built entirely out of Education, Ethics/ Society, and Environment items. This is a much sharper version of the “Technology sits on the periphery” finding from the Cytoscape degree view: it is not just that some Technology items are weakly connected, it is that none of them are strong enough to break into the class’s twenty tightest bonds at all.

Most of the strongest bonds cross themes, not stay within them. Only 6 of these 20 pairs connect two items from the same theme (E11 x E15, E14 x E15, E10 x E15, V09 x V13, V13 x V14, V09 x V14); the other 14, 70% of the list, link two different themes. The single tightest bonds in this class are not “people who like one Environment item like other Environment items,” they cut across topic boundaries entirely. A good concrete example is rank 7, S05 (“individuals should have greater control over how their personal data are collected and used”) paired with V13 (“companies should be held accountable for the environmental impacts of their activities”). These two statements share no obvious subject matter, one is about digital privacy and the other about corporate environmental harm, yet they are endorsed together almost as strongly as two items about the exact same topic. The natural reading is that both are really asking the same underlying question in different clothes: should powerful institutions (tech platforms, corporations) be held accountable to individuals and the public. It is also worth noticing that E11, an Education item, is itself explicitly about technology (“update curricula more frequently to match emerging technologies”), yet it behaves like a member of the Ethics/Environment package (pairing with S05, S11, V09, V13) rather than clustering with the Technology theme it is adjacent to in subject matter. The T/E/S/V labels describe where a statement sits in the survey form, not necessarily which latent belief cluster it actually belongs to.

Analysis and Visualizations

The figures above already carry the bulk of the analysis, step by step, alongside the algorithms that produced them. Pulling the threads together:

- The agent network shows a shared, moderately agreeable backbone rather than two opposed camps: giant component covering 80.5% of respondents, moderate Louvain modularity ($Q = 0.275$), and a lopsided Girvan-Newman cut (64/4/2).
- The two large Louvain communities differ mainly in intensity of agreement, not direction, and Education is the lowest-scoring theme in both.
- The attitude network shows a coherent, tightly co-endorsed core of Environment and Ethics/Society statements (plus a few forward-looking Education items), with several Technology and classroom-process statements sitting on the periphery or fully isolated.
- Response variance and structural-importance rankings both point to Education items (E04, E03, E01) and specific Technology items (T02, T14) as the statements that most divide, and most organise, opinion-neighbours in this class.
- T14 (social media and polarization) is a uniquely important bridge item: by far the highest betweenness centrality in the attitude network, while several other central items (V13, E15, S05, E11) are important because they sit at the well-connected heart of the network rather than acting as bridges.

Results and Discussion

What we learned about the survey and the class

1. A shared, moderately-agreeable backbone, not two hostile camps. After filtering (dropping 9 incomplete respondents) and a literature-motivated agreement threshold ($\theta = 41.5$, giant component covering 80.5% of respondents), most of the class sits in one well-connected agreement network. Girvan-Newman’s lopsided 64/4/2 split and Louvain’s moderate modularity ($Q = 0.275$) both point away from a bipolar divide.
2. Two large opinion neighbourhoods that differ in intensity, not direction. The two big Louvain communities (37 and 29 members) are positive on every theme; one is simply more strongly and consistently positive than the other, especially on Ethics/Society and Environment. Education is the lowest-scoring theme in both, a shared area of muted enthusiasm rather than a dividing line.
3. Education, and specific AI-in-education items, are the real fault lines. High response variance (E04, E03, E02), high structural importance under the shuffle test (E04, E01, T02), and peripheral position in the attitude network together point to classroom-practice and generative-AI-in-learning statements as where opinion-neighbours genuinely diverge.
4. A coherent “responsible citizen” belief package. The attitude network’s dense core, visible in the Cytoscape degree view and the eigenvector ranking alike, bundles environmental responsibility, social responsibility, personal-data control, and a handful of forward-looking education statements into one tightly co-endorsed package, distinct from a looser set of Technology and classroom-process statements.
5. T14 (“social media algorithms cause polarization”) is a uniquely important bridge statement, with by far the highest betweenness centrality in the attitude network, and independently ranking among the top items in the shuffle-based importance test.

A note on a possible confound

While preparing the data, we noticed that missing answers are not spread evenly across the four themes: Technology, presented first, has the fewest missing cells, and Environment, presented last, has the most (see Dataset Documentation). Environment and Ethics/Society, the two last-presented themes, are also the two themes with the highest mean scores and the most internally consistent (least divided) communities.

Both a real-attitude explanation and a survey-fatigue explanation are consistent with this pattern. Environment and Ethics/Society may simply be less controversial topics than Technology and Education. It is also possible that respondents answer less carefully as a long Likert survey goes on, a well-documented tendency in survey research called

satisficing (Krosnick, 1991), which would make the later themes look more uniformly agreeable than respondents' true opinions. Because item theme and item position are perfectly tied together in this survey (every Technology item comes before every Education item, and so on), our existing pipeline cannot fully separate the two explanations, and we flag this as a limitation rather than a settled finding: the strength and uniformity of agreement on Environment and Ethics/Society items should be read with some caution.

How the network approach helped

A table of column means would have correctly reported the class's overall Agree skew and theme-by-theme averages. It would have missed the backbone-versus-fringe geometry of the class at a principled agreement threshold, the existence of two opinion neighbourhoods differing in intensity rather than direction, the distinction between an item being contested and an item being community-organising, and the co-endorsement packaging of statements into a coherent worldview cluster. The MacCarron/Dinkelberg pipeline turned a high-dimensional Likert table into relational structure we could visualise, measure, and interrogate, which is what let us move from “the class mostly agrees” to a more specific picture: the class shares a broad societal-and-environmental core, with classroom practice and generative-AI-in-education as the areas of real, structurally meaningful disagreement.

Individual Contribution

Member	Contribution
Kushal Balabhadruni	Dataset documentation, Likert encoding and missingness rules, notebook 01 network construction
Rohit Jeswanth	Similarity thresholding, GraphML / theme-slice exports, community detection and item-importance analysis in notebook 02
Vishak Kashyap K	Figure theme and Cytoscape views, report narrative and assembly, repository organization

References

1. MacCarron, P., Maher, P. J., and Quayle, M. (2020). Identifying opinion-based groups from survey data: a bipartite network approach. arXiv:2012.11392.
2. Dinkelberg, A., MacCarron, P., Maher, P. J., and Quayle, M. (2021). Detect opinion-based groups and reveal polarisation in survey data. arXiv:2104.14427.
3. Maher, P. J., MacCarron, P., and Quayle, M. (2020). Mapping public health responses with attitude networks: the emergence of opinion-based groups in the UK’s early COVID-19 response phase. *British Journal of Social Psychology*.
4. Krosnick, J. A. (1991). Response strategies for coping with the cognitive demands of attitude measures in surveys. *Applied Cognitive Psychology*, 5(3), 213-236.